

Discussion 1: Hand with Reflecting Sphere

Discussion Topic:

Computer vision is a field of computer science that aims at enabling computers to process and identify images and videos in the same way that human vision does. The main objective is to describe a real-world scene in one or more images.

Take a look at the image [Hand with Reflecting Sphere](#) by M. C. Escher from 1935. Write a post of no less than a paragraph addressing the following question:

- What is in this image and why might this question be hard to answer with computer vision?

My Post:

Hello Class,

The “Hand with Reflecting Sphere” lithograph by M. C. Escher (1935) is not only a great hand-drawn artwork but also an impressive illustration of perspective distortion and reflection caused by the curvature of a spherical mirror. It is a complex scene where a human hand holds up, in stand like posture, a ball-shaped spherical mirror. The surface of the mirror reflects a sitting man in a room with chairs, his hand holding it, shelves containing books, a window, and other furniture. Additionally, the convex curvature of the mirror distorts the dimensions of the room and the object being reflected in it.

A computer vision system may identify the objects but would not necessarily be capable of explaining or understanding what is happening in an image. For instance, in the scene portrayed in the "Hand with Reflecting Sphere" image, such a system will not necessarily understand that the room and the object in the room are being reflected and that their shapes are distorted due to the curved surface of the spherical mirror; it would not necessarily understand that it is the phenomenon of light being reflected. An advanced vision system may identify a hand holding a room containing a man, furniture, books, a window, etc. A basic vision system may identify the objects within the image but will have no understanding of how the objects are related to one another or why they appear the way they do. In other words, the systems may be capable of identifying objects, but they will not be capable of explaining in any meaningful way the composition of the image or what is happening in the image.

This gap between object identification and actual understanding can be explained by how computer vision systems process images using raw pixel data. Krishna (2017) explains that what a computer system interprets is based on the image's numeric pixel values, and that converting those numerical values into meaningful information is one of the most challenging aspects of computer vision. In other words, for a computer vision system to understand a scene, it must convert raw pixel values into meaningful information rather than simply detect changes in

brightness or recognize familiar shapes. This is not an easy thing to do for a computing system, as computer systems cannot generalize and contextualize a scene or composition at the same level as humans.

In the “Hand with Reflecting Sphere” image, a computer system may detect the hand supporting the ball-shaped spherical mirror and its reflection on it as two different hands. Yang et al. (2019) explain that mirrors can confuse computer vision systems because the objects they reflect resemble physical objects in a space, making it difficult for the systems to distinguish the objects from their reflections. Additionally, identifying a chair in an image does not automatically mean that the system knows where the chair is located in a scene, or that the object is a reflection of a chair located in a different physical location within the scene.

Furthermore, the distortion created by the curvature of the mirror can also be difficult for a computer vision system to interpret, even for a system trained to recognize reflections, because the object’s reflection no longer resembles the dimensional proportions of the object being reflected, confusing the system and causing it to misinterpret the reflection as a different object rather than a distorted reflection of an object.

Overall, Escher's image illustrates why identifying individual objects is only part of computer vision. To understand what is actually happening in the image, a system must also interpret the reflection, the distortion caused by the spherical mirror, and the spatial relationships between the objects being reflected. This is what makes scene understanding considerably more difficult than simply recognizing objects.

-Alex

References:

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